Assessing ears

Weber: Chapter 17

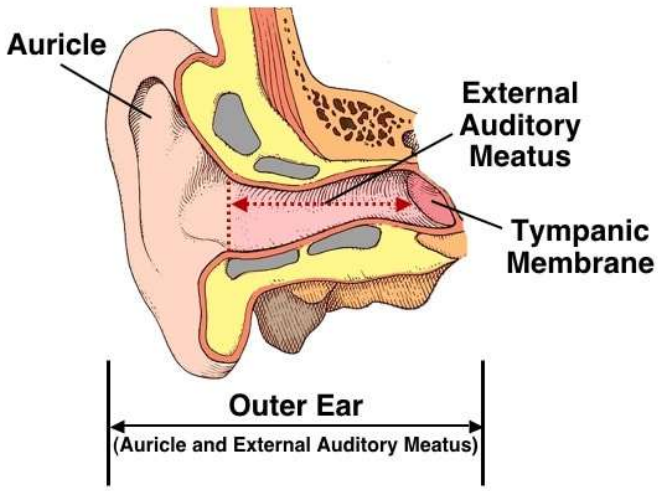
NURS3120: Health Assessment and Promotion

Objectives

- 1. Describe the structures and functions of the ears
- 2. Discuss the risk factors for hearing loss and ways to reduce one's risks
- 3. Differentiate between normal and abnormal findings of the ears and hearing
- 4. Analyze the client's interview and physical assessment data related to the ears and hearing to formulate valid clinical judgments

External ear

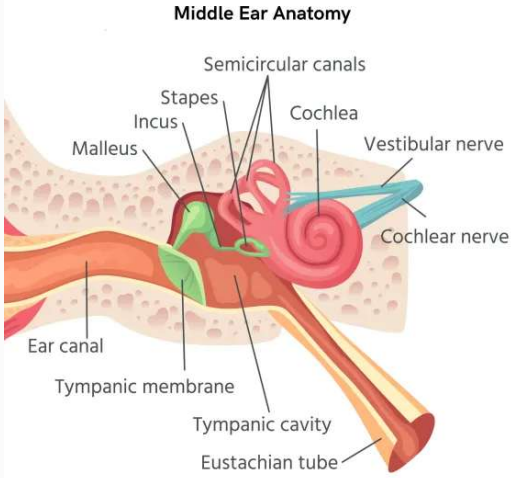
- Auricle (Pinna)
- External auditory canal
- Tympanic membrane



The pinna is the portion of the external ear visible without any tools. It is composed of a thin plate of yellow elastic cartilage. It is shaped with hollows, furrows, and ridges that form an irregular funnel to conduct sound waves into the external auditory canal.

The external auditory canal is S shaped in the adult. The outer part of the canal curves up and back; the inner part of the canal curves down and forward. Modified sweat glands in the external ear canal secret cerumen, ear wax. Cerumen has bacteriostatic properties; it is sticky and the ear's defense against foreign bodies. Ear wax keeps the tympanic membrane soft.

The tympanic membrane Also known as…….. Eardrum. It is translucent, pearly gray appearance and serves as a partition stretched across the inner end of the auditory canal, separating it from the middle ear.



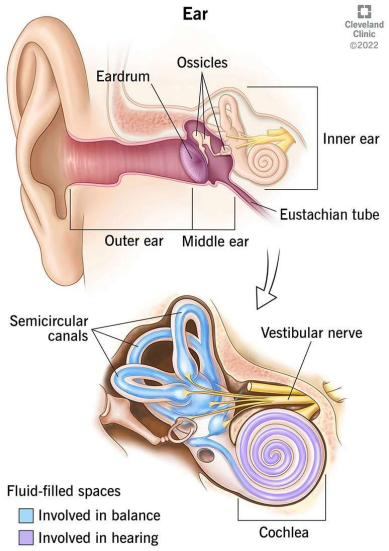
Middle ear

- Air filled chamber in the temporal bone
- Auditory ossicles:
  - Malleus
  - Incus
  - Stapes

The middle ear or tympanic cavity is a small air-filled chamber in the temporal bone. The middle ear contains three auditory ossicles the malleus, the incus, and the stapes. These tiny bones are responsible for transmitting sound waves from the eardrum to the inner ear through the oval window. Air pressure is equalized on both sides of the tympanic membrane by means of the eustachian tube which connect sthe middle ear to the nasopharynx.

Inner ear

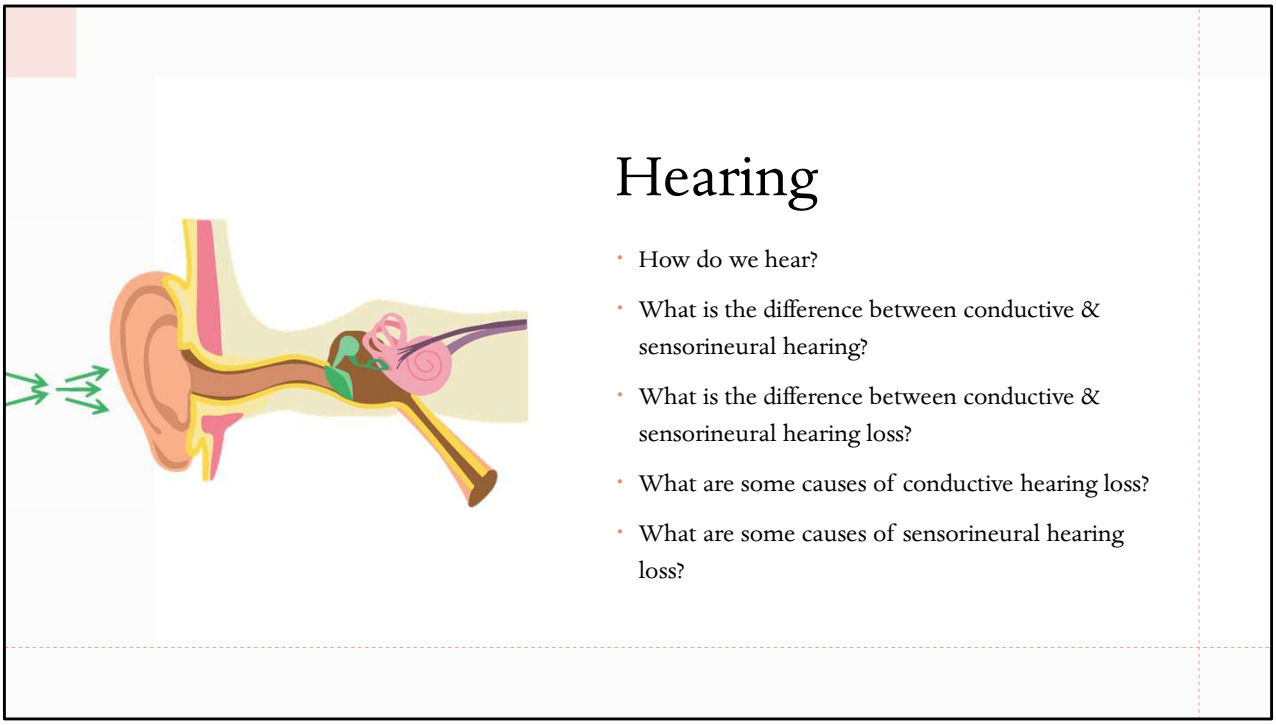
- Fluid filled & made up of:
  - Boney labyrinth
    - Cochlea
    - Vestibule
    - Semicircular canals
  - Inner membranous labyrinth



The inner ear, or labyrinth, is fluid-filled and made up of the bony labyrinth and an inner membranous labyrinth. The bony labyrinth has three parts: the cochlea, the vestibule, and the semicircular canals.

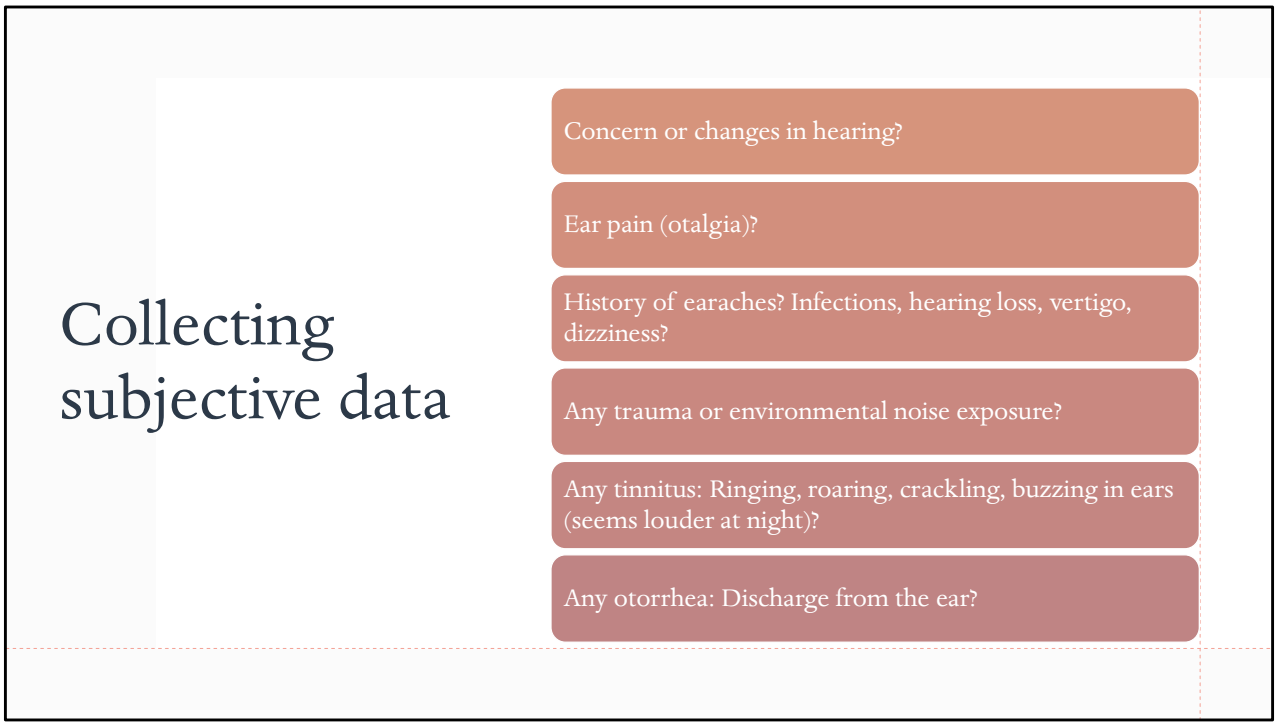
The inner cochlear duct contains the spiral organ of Corti, which is the sensory organ for hearing. Sensory receptors, located in the vestibule and in the membranous semicircular canals, sense position and head movements to help maintain both static and dynamic equilibrium.

Nerve fibers from these areas form the vestibular nerve, which connects with the cochlear nerve to form cranial nerve VIII (acoustic or vestibulocochlear nerve).



Sound vibrations travel through air and are collected by and funneled through the external ear causing the eardrum to vibrate. Sound waves are then transmitted through auditory ossicles as the vibration of the eardrum causes the malleus, the incus, and the stapes to vibrate. As the stapes vibrates at the oval window, the sound waves are passed to the fluid in the inner ear. The movement of this fluid stimulates the hair cells of the spiral organ of Corti and initiates the nerve impulses that travel to the brain by way of the acoustic nerve.

The transmission of sound waves through the external and middle ear is referred to as conductive hearing and the transmission of sound waves in the inner ear is referred to as sensorineural hearing.
Conductive hearing loss would be related to the dysfunction of the external or middle ear (impacted earwax, otitis media, foreign object, perforated eardrum)
Sensorineural hearing loss would be related to dysfunction of the inner ear (cranial nerve 8 or temporal lobe of the brain, or the organ of Corti)-
Sensorineural hearing loss is the most common type of permanent hearing loss. It decreases one's ability to hear faint sounds. Even loud speech may be muffled. (Aging, virus, ototoxic drugs, genetic hearing loss, head trauma, loud noise exposure, Meniere disease, autoimmune inner ear disease)



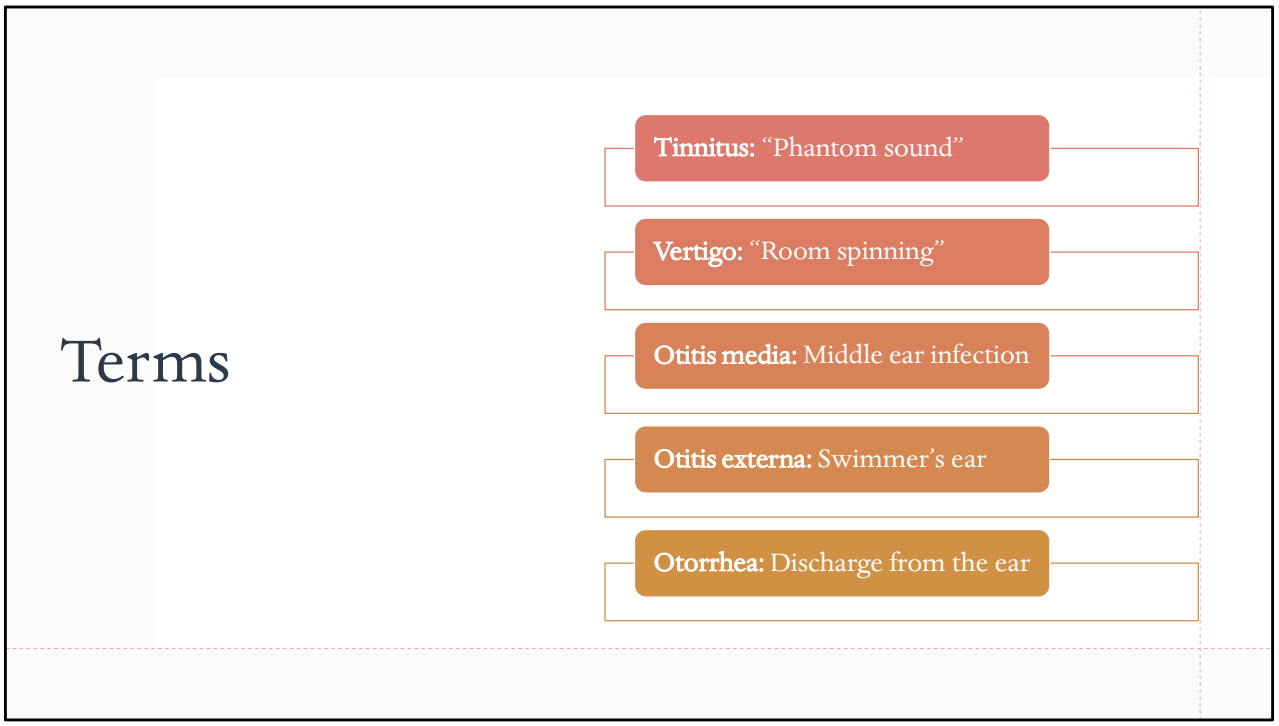
Concern for hearing loss? Sometimes asking a family member about any hearing loss.
Ear Pain: Always use COLDSPA. Pain can be caused by infections, cerumen blockage, sinus infections, teeth or gum problems.

Any trauma or environmental noise exposure? Ringing, Roaring or crackling in your ears? This could be caused by trauma or environmental noise exposure. Some adults who work at airports, construction, and so forth complain of tinnitus.

Vertigo: True spinning motion may be associated with an inner ear problem. Subjective vertigo is when a client feels like they are spinning around, and objective vertigo is when a client feels that the room is spinning around them.

Vertigo versus dizziness: Vertigo is the feeling of movement and dizziness is a feeling like they are about to pass out or a sensation of floating.

Otorrhea: Usually indicated infection. History of repeated infections can affect the tympanic membrane (possible rupture) and hearing problems.



Tinnitus – occurs with cerumen impaction, middle ear infection, HTN, hearing loss. Can be related to environmental noise, many medications have ototoxic properties that may cause tinnitus or hearing loss: aspirin, aminoglycosides (gentamycin, tobramycin), furosemide, naproxen, vancomycin
Can be very debilitating.

Otitis media – middle ear infection, very common in children

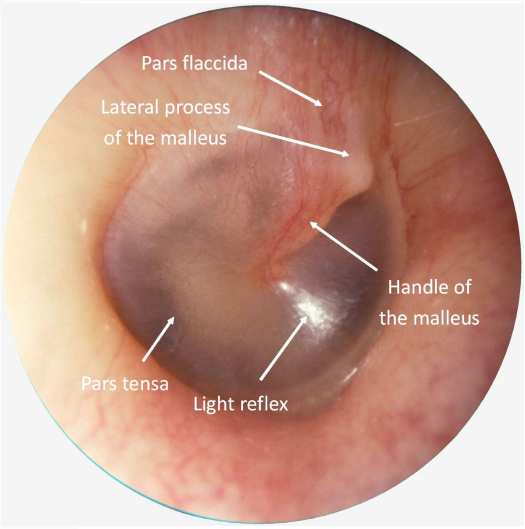
Otitis externa – inflammation of the ear canal. Can be fungal infection from moisture "Swimmers ear." Can also be from inserting something in the ear canal like a bobby pin/key/fingernail to clean or scratch. Otitis externa causes pain when the pinna is moved and typically has a foul-smelling cerumen.



Inspection & palpation of the external ear

- Auricle
- Tragus
- Lobule

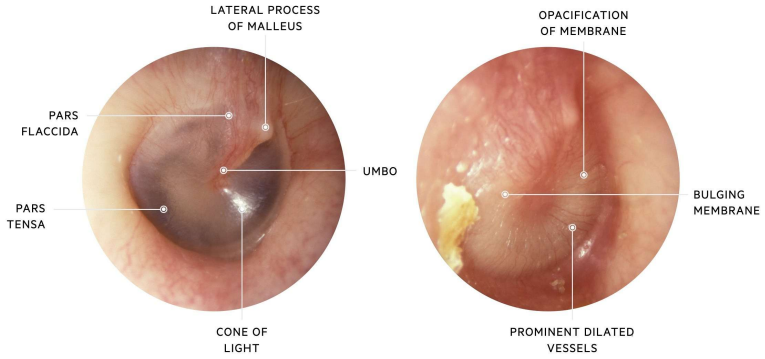
Inspect the ear: Ears are equal, note shape, size, position. Skin should be smooth with no lesions, lumps or modules. Color is consistent with facial color. Abnormal findings suggest various disorders. Including: Enlarged preauricular and postauricular lymph nodes/infection. Tophi are nodules containing uric acid located on the helix could be gout. Ulcerated, crusted nodules that bleed could be skin cancer.
Be mindful of earrings and piercings. Some people have the tragus or daith pierced for migraines.
Palpate the auricle and mastoid process. Normally the auricle, tragus and mastoid process are not tender. Abnormal finding is a painful auricle or tragus that is associated with otitis externa swimmer's ear) or postauricular cyst. Tenderness over the mastoid process suggests mastoiditis and tenderness behind the ear may occur with otitis media.



Inspection of the tympanic membrane

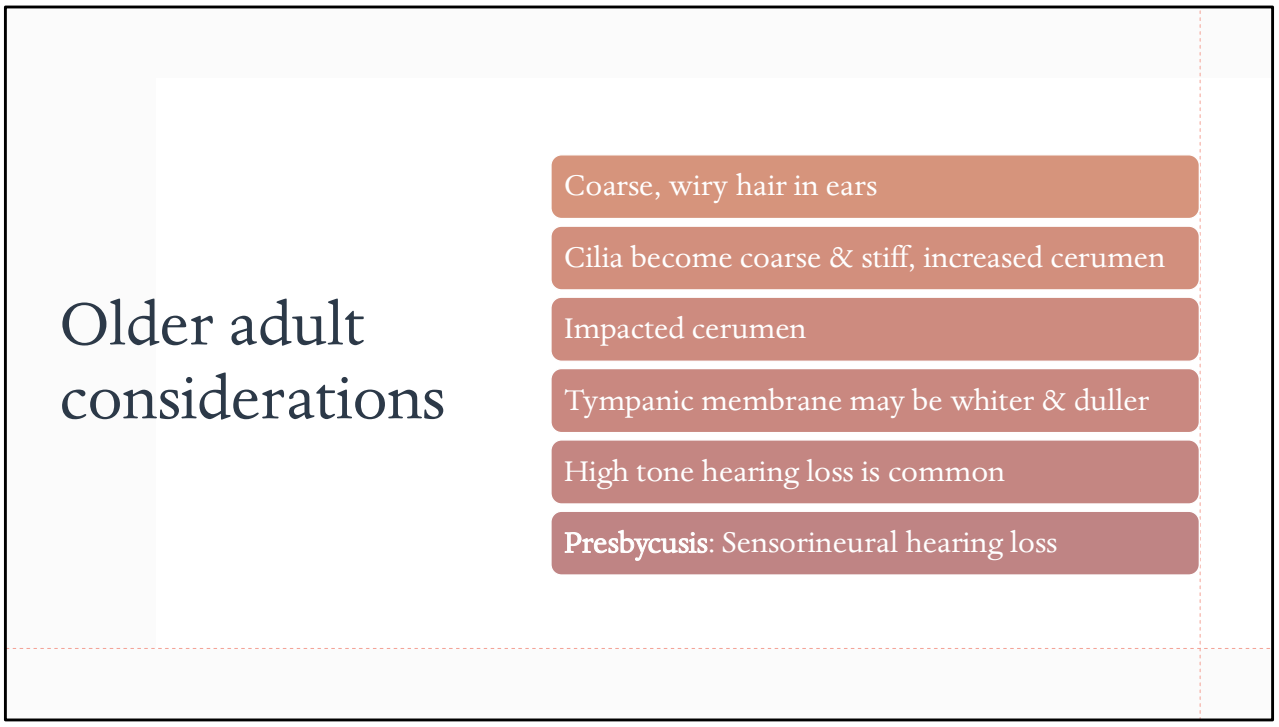
- Normal:
  - Pearly grey
  - Shiny
  - Translucent
  - No bulging
  - No retraction

Tympanic membrane- Pearly gray in color, shiny, translucent. Should appear flat and slightly pulled in the center.
Abnormal findings:
Red, bulging eardrum and distorted, diminished, or absent light reflex—acute otitis media
Yellowish, bulging membrane with bubbles behind—serous otitis media
Bluish or dark red color—blood behind the eardrum from skull trauma
White spots—scarring from infection
Perforations—trauma from infection
TIP: Please review the abnormalities of the tympanic membrane on 353.



Normal vs. abnormal

- Normal tympanic membrane on left
- Acute otitis media on right



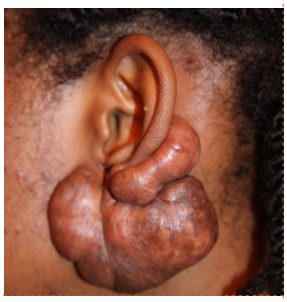
Presbycusis- gradual onset, caused by nerve degeneration in the inner ear, high-frequency tone loss, harder to hear consonants than vowels, words sound garbled. Female voices are harder to hear. About one in three people in the US age 65-74 has hearing loss, and almost one half of those older than age 75 have difficulty hearing. Is common in older clients 50 and older. Normal part of aging, but still an ABNORMAL finding.

Cerumen production decreases, leading to dryness and tendency toward accumulation.

Abnormal findings of the external ear







Malignant lesion, tophi, keloid scars

Health promotion and disease prevention: Hearing loss

- Hearing affects all interpersonal communication
- Safety issues:
  - Difficulty following doctor's orders
  - Respond to warnings
  - Hearing doorbells and alarms
- Can affect childhood development
- May have genetic or nongenetic cause

Healthy People 2030 goal: Reduce the prevalence and severity of disorders of hearing and balance; smell and taste; and voice, speech, and language

Congenital hearing loss is associated with maternal rubella, syphilis, or certain other infections during pregnancy; low birth weight; birth asphyxia (a lack of oxygen at the time of birth); inappropriate use of particular drugs during pregnancy, such as aminoglycosides, cytotoxic drugs, antimalarial drugs, and diuretics; and severe jaundice in the neonatal period, which can damage the hearing nerve in a newborn infant.

Acquired causes include infectious diseases such as meningitis, measles, and mumps; chronic ear infections; collection of fluid in the ear (otitis media); use of particular drugs, such as some antibiotic and antimalarial medicines; injury to the head or ear; excessive noise, including occupational noise such as that from machinery and explosions and recreational noise such as that from personal audio devices, concerts, nightclubs, bars, and sporting events; aging, in particular due to degeneration of sensory cells; and wax or foreign bodies blocking the ear canal.

Hearing loss

- Newborns should be screened for hearing loss by one month of age, preferably before leaving the hospital
- Risks: Aging, heredity, occupational noise, recreational noise, ototoxic medications, illness (especially w/high fever)
- Client education: Avoid loud sound exposure or wear hearing protectors, immunize children against childhood diseases, being immunized for rubella before pregnancy or if childbearing age, avoid use of ototoxic medications, avoid putting foreign bodies in ears





Questions?

